FIT2099 Assignment 3 Design Rationale

Requirement	Author
REQ 1: Second Spring	Poyee Tan 36641162
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REQ 5: The Wanderer (API, HD)	Jingyu Zhang, Yuanshu Liu, Poyee Tan

Poyee Tan 36641162 – REQ 1: Second Spring

Class diagram: `docs/A3_UML/A3_REQ1_UML.drawio`. Feature proposal: `feature-proposal.md`.

REQ1 makes plants alive. Every plant has its own age, which starts at 0 and goes up by 1 each turn. Each turn the plant tries its abilities in priority order (grow, spread, or drop produce) and stops at the first one that succeeds, so a plant does at most one thing per turn. Only tilled Soil accepts new plants. Every effect changes the map, never only prints text.

Class	Abstraction	Effect	Game state it changes
`Plant`	`Ground`, `Regressible`	Ages each turn, tries its abilities, can be set back one stage	Its own age; the ground on its tile
`PeriodicAbility`	`PlantAbility`	Acts every N turns of age with a given chance	— (template only)
`Grow`	`PeriodicAbility`, uses `Growing`	Replaces the plant with a new instance of its next stage	The ground on the tile
`SelfReplicate`	`PeriodicAbility`, uses `Spreading`	Plants a new plant on a random adjacent Soil tile	The ground on a nearby tile
`Produce`	`PeriodicAbility`, uses `Producing`	Drops a new item on a random adjacent tile	Items on a nearby tile
`StrawberrySeedling`, `StrawberrySprout`, `CauliflowerSprout`, `MelonSprout`	`Plant`	The young stages, which grow and (sprouts) spread	—
`Plantable`, `PlantableTile`	interface, `TileCondition`	Say which ground a plant may be planted on	—
`GrowthPausedStatus`	`TimedStatus`	Stops a plant's abilities for a few turns	Whether the plant acts

Plant	Abilities, highest priority first	Capabilities	Regresses to
`StrawberrySeedling`	Grow (every 5 turns, 50%)	`Growing`	Dirt
`StrawberrySprout`	Grow (10, 50%), SelfReplicate (5)	`Growing`, `Spreading`	`StrawberrySeedling`
`CauliflowerSprout`	Grow (15, 25%)	`Growing`	Dirt
`MelonSprout`	Grow (15, 25%), SelfReplicate (10)	`Growing`, `Spreading`	Dirt
`Strawberry`	Produce (5)	`Producing`	`StrawberrySprout`
`Melon`	SelfReplicate (20)	`Spreading`	`MelonSprout`
`Tomato`	Produce (7), SelfReplicate (20, 10%)	`Producing`, `Spreading`	Dirt
`Parsnip`	Produce (15)	`Producing`	Dirt
`Cauliflower`	none	none	`CauliflowerSprout`

Each decision compares rejected and chosen designs, with pros, cons and a future example. Sections I and K list every place a reader may see as a weakness, and say why we chose it.

A. One class per growth stage

Rejected 1: one class per crop with a `stage` field and a `switch` that picks the character, abilities and rules.

Pros	Cons
KISS: one class holds a crop's whole life.	OCP: a new stage edits every `switch` (the "switch statement" smell).
No new classes.	SRP: one class owns several display characters, ability lists and harvest rules.

Pros	Cons
	A sprout would still offer watering and harvesting, guarded by `if (stage == SPROUT)` checks.

Rejected 2: one class per crop holding a separate stage object (State pattern) that it swaps as it grows.

Pros	Cons
The crop keeps one identity on the tile.	The spec says growing creates "a brand new plant with age 0"; a swapped state would have to reset the age by hand (Connascence of Execution).
	The tile object would have to change its character, name and allowed actions at run time.

Chosen: one `Plant` subclass per stage. `Plant` is a new abstract class between `Ground` and `Crop`, and `Crop` now extends it. Growing replaces the ground with a new object, so the new stage starts at age 0 with no reset code. The young stages extend `Plant`, not `Crop`: as `Crop`s they would inherit watering and harvesting they must refuse.

Pros	Cons
OCP: a new stage is one new class. No existing stage changes.	Four small classes that mostly hold constants.
LSP: every `Crop` really can be watered and harvested; no sprout refuses a `Crop` method.	Stages name each other, e.g. `StrawberrySprout.nextStage()` returns `new Strawberry(...)` (Connascence of Name, kept inside `game.crops`).
SRP: each class has one character, one ability list and one set of rules.	

Package placement. The young stages live in `game.crops`, beside the crops they grow into, not in `game.plants`. Otherwise `game.plants` would import `game.crops`, which imports `game.plants`: a cycle (Acyclic Dependencies Principle). `game.plants` now depends on no crop.

Foresight: a Pumpkin Sprout that grows into a Pumpkin is one new class; no other class changes except `StackQueueValleyWorld`, which places it.

B. Plant behaviour as a list of abilities

Rejected: each plant overrides `tick` with its own if-else.

Pros	Cons
Each plant's rules sit in one place.	DRY: "every N turns, roll a chance, stop at the first success" is copied into nine classes. A bug fix needs nine edits (Connascence of Algorithm).
	Easy to break the "one thing per turn" rule in one plant without noticing.

Chosen: `Plant` holds a `List<PlantAbility>` in priority order, passed to `super(...)`. `PlantAbility` has one method, `tryPerform(plant, location)`, which returns a report, or `""` when nothing happened. `Plant.tick` ages the plant, then `attemptGrowth` tries the abilities from the top and stops at the first non-empty report. `PeriodicAbility` implements the timing once: its `tryPerform` is `final` (Template Method). It checks `age % interval`, rolls the chance, then calls the abstract `perform`. `Grow`, `SelfReplicate` and `Produce` write only their effect.

Pros	Cons
DRY: the timing rule and the priority loop each exist once.	A reader opens two classes (the plant and the ability) to see one behaviour.
OCP: a new ability, e.g. "wither", is one new class.	Priority is the order of the constructor arguments (Connascence of Position). Each plant's class comment lists its abilities in priority order.
DIP: `Plant` depends only on `PlantAbility`.	`final tryPerform` stops an ability from changing the timing; an ability with different timing would implement `PlantAbility` directly.

Pros	Cons
SRP: `Plant` decides when to act; each ability decides what happens.	

`PeriodicAbility` rejects an interval below 1 or a chance outside 0 to 100 in its constructor; an interval of 0 used to crash on `age % 0`. `attemptGrowth` is `public` and does not age the plant, so REQ4's Bounty Enchantment reuses it to force an extra growth attempt instead of copying the loop (DRY across requirements).

Foresight: a Rose that withers after 30 turns is one new `PeriodicAbility` subclass placed first in its list.

C. Plant capabilities: `Growing`, `Spreading`, `Producing`

Each ability must ask the plant what to create: `Grow` needs the next stage, `SelfReplicate` the offspring, `Produce` the item. The plant knows this, and already holds the `RandomUtils` and `AreaFactory` the new object needs.

Option	Pros	Cons
A `Supplier` per ability, e.g. `new Grow(..., StrawberrySprout::new)` (our first version)	The ability is given exactly what it creates.	The team removed method references and functional style everywhere after A2 feedback. The supplier also needed the plant's `RandomUtils` and `AreaFactory`.
Three methods on `Plant` whose defaults throw `UnsupportedOperationException` (our draft)	One abstract class; each plant overrides only the methods its abilities call.	ISP, Refused Bequest: Parsnip only produces, yet carried `nextStage()` and `offspring()` that could only throw. The Javadoc had to say "the default throws".
An abstract class per capability	Shared code could live in it.	Java has single inheritance: `Strawberry` already extends `Crop`, so it could not also extend a `ProducingPlant`.
One single-method interface per capability (chosen)	See below.	See below.

Chosen: `Growing` (`nextStage()`), `Spreading` (`offspring()`) and `Producing` (`produce()`). Each plant implements only what it can do. An ability finds its capability with `location.getGroundAs(Growing.class)` and does nothing if it is missing, the same way pests find `Regressible`.

Pros	Cons
ISP: no plant carries a method it cannot honour, and nothing throws.	Three more interfaces.
No `instanceof`: `getGroundAs` is the engine's capability lookup.	A plant given `Grow` but not `Growing` silently never grows instead of failing. `PlantCapabilityTest` and the per-plant tests catch this.
Consistent with `Regressible` and `Plantable`, so a reader learns one pattern.	The capability methods must be `public`, as interface methods are.
DIP: abilities depend on three small interfaces, not on `Plant` subclasses.	

Under Design by Contract, the throwing draft was not strictly an LSP violation (no override weakened a postcondition, and no path reached the throw), so we describe it as an ISP problem.

Foresight: a Sunflower that drops seeds implements `Producing` and is given `Produce`; no ability changes.

D. Counting turns

Option	Pros	Cons
The engine's `Status` countdown	Already exists.	Statuses tick and expire independently, so they cannot express "stop at the first success".
A global turn counter	One number for the whole game.	The engine has none; adding one would need a static field, which A2 feedback penalised.

Option	Pros	Cons
The engine's `Behaviour<Plant, String>` as the ability type	No new interface.	Its contract returns `null` for "nothing", and it is shared with actors; our contract is plant-specific and returns `""`.
`Plant` keeps its own `age` (chosen)	Each plant counts from 0 when placed, as the spec says; no setter, so only the plant changes it.	A plant spread right of or below its parent ages once in its first turn, because the engine ticks tiles row by row.

Randomness comes from a constructor-injected `RandomUtils`, so every test passes a `FixedRandom` and is deterministic.

E. Where plants can grow

Rejected: `instanceof Soil` wherever something is planted.

Chosen: a `Plantable` interface with `plant(ground, location)`, and a `PlantableTile` condition for `Area.filter`. `Soil` implements it. `plant` takes a `Ground`, not a `Plant`, so `game.positions` does not depend on `game.plants` (no package cycle).

Pros	Cons
OCP: a new plantable ground, e.g. fertilised soil, only implements the interface.	One more interface and one more condition class.
DRY: `SelfReplicate`, the Merchant's `Crop.plantNextTo` and REQ5's Sowing omen all plant through it, so they follow one rule.	
Reused by REQ3: the mole and the snail find bare Soil through it.	

In Assignment 2 the Merchant could plant on Dirt; now it plants only on Soil, as the spec requires.

Foresight: a Greenhouse floor that accepts plants implements `Plantable`; every planting path accepts it at once.

F. Regressing, for REQ3 pests

Rejected: pests call `Crop` methods directly.

Pros	Cons
No new interface.	Pests would name every crop and stage; a new stage would edit every pest (OCP).

Chosen: `Plant` implements `Regressible`, whose `regress` replaces the plant with `previousStage()`. By default a regressed plant dies and leaves `Dirt`, matching A2's `Crop.removeCrop`; later stages override `previousStage()` to return their earlier stage.

Pros	Cons
DIP: pests depend on `Regressible`, so `game.pests` does not import `game.crops`.	A pest cannot treat one crop differently.
OCP: a new plant is eaten by every pest with no change.	`previousStage()` is `protected`, so classes in the same package could call it. None does.

G. Pausing growth, for REQ5

Option	Pros	Cons
The status removes the plant's abilities and puts them back (push)	The plant needs no check.	The status would reach into `Plant`'s private list and must restore it exactly (Connascence of Identity).
The plant checks for `GrowthPausedStatus` (pull, chosen)	Deciding whether to act stays the plant's job (SRP); the status only counts down.	`applyEffect` is empty on purpose, which a reader may see as a do-nothing override; a comment says why.

While paused, the plant still ages. REQ5's Frost omen applies the status; `TimedStatus` (A2) supplies the countdown, so no timer code is repeated.

H. Printing without a Display parameter

`Ground.tick` returns nothing, so a plant prints its own report.

Option	Pros	Cons
`private static final Display` (our draft)	One shared object.	Shared static state, which A2 feedback penalised.
A `Display` injected through the constructor	Strict DIP.	Nine constructors, every `nextStage()` and `offspring()`, `StackQueueValleyWorld` and every test gain a parameter, yet `Display` is the engine's console writer with no alternative to swap in.
Return the report and let someone else print it	No `Display` in `Plant`.	Nothing calls `Ground.tick` except the engine, which ignores any result.
`private final Display` per plant (chosen)	No static state; KISS.	Each plant creates a `Display`, a small DIP gap.

The collaborators that do vary, `RandomUtils` and `AreaFactory`, are injected.

I. Visibility, `final`, `protected` and `static`

Choice	Where	Why	Cost
Every field is `private`	All REQ1 classes	Implementation details stay inside (A2 feedback).	Each plant keeps its own `RandomUtils` and `AreaFactory` to pass to the plants it creates.
Constants are `private static final`	e.g. `GROW_INTERVAL`, `PRICE`	A name explains each number; Java does not allow instance fields in `super(...)`.	Tests repeat the numbers.
No static methods and no mutable static fields	—	A2 feedback penalised static methods.	—
`protected` constructors	`Plant`, `PeriodicAbility`	They are abstract; only subclasses construct them.	—
`protected` hooks	`perform`, `previousStage`	Subclasses override them; callers must go through the template.	Same-package classes could call them; none does.
`final` methods	`tryPerform`, `plantNextTo`, `noRoomToPlant`	Fix the shared rule so a subclass cannot change it.	A plant needing different timing implements `PlantAbility` directly.
`public` capability methods	`nextStage`, `offspring`, `produce`	Interface methods must be public.	Any class could ask a plant for its next stage; only the abilities do.
Classes are not `final`	e.g. `Grow`, `Soil`	Open for extension (OCP).	Private fields and `final` templates limit the damage.

J. Changes to Assignment 2

Change	Why	Cost
`Crop` extends `Plant` instead of `Ground`	Every crop gets an age, abilities and `Regressible`.	`Crop` constructors take abilities. Watering and harvesting are unchanged.
`Unplanted` checks for any `Plant`	Watering a Cauliflower used to flood sprouts into puddles.	—

Change	Why	Cost
The Merchant plants through `PlantableTile`	The spec says only Soil accepts plants.	A Merchant surrounded by Dirt cannot sell crops; the message says so.
`RandomUtils` and `AreaFactory` are injected instead of static	A2 feedback; tests pass `FixedRandom`.	Every plant constructor has two more parameters ("tramp data" smell).
Fixed farm layout with one of every plant	Every plant is visible when the game starts.	—
Unique characters: Crow `b`, Cauliflower `K`	The spec gives `v` to the Strawberry Sprout, and `C` is the Cow.	Players of A2 see new characters.

K. Known minor violations and why we accept them

Violation or cost	Why it is acceptable
`SelfReplicate` and `Crop.plantNextTo` repeat the two lines that find an adjacent Plantable tile and plant on it. DRY is not fully met.	They build different messages. A third use would be the signal to extract it.
`Crop` also builds the Merchant's "no room to plant" message, mixing trading into a ground (SRP).	It keeps one wording for both purchasable crops, Tomato and Parsnip.
Each plant creates its own `Display` (Section H).	Injecting it adds parameters everywhere without making anything replaceable (KISS).
A plant given an ability without the matching capability does nothing (Section C).	Safer than a crash; tests check every plant's abilities.
The engine ticks tiles row by row, so a plant spread right of or below its parent ages once in its first turn.	Engine behaviour; it shifts timing by one turn and breaks no rule.
`StackQueueValleyWorld` registers the map characters with method references, `Dirt::new` (base code) and `Soil::new` (REQ1), although the team avoids functional style.	The engine's `DefaultGroundCreator.registerGround(char, Supplier<? extends Ground>)` needs a `Supplier`, and the engine cannot be changed. A named class that implements `Supplier` for each ground would be a class used once, only to avoid one line (KISS). These are the only two method references in the game code.

Testing

`GrowTest`, `SelfReplicateTest`, `ProduceTest`, `PlantStateTest`, `PeriodicAbilityTest` and `PlantCapabilityTest` cover every plant at the right age, one turn early, with a failed chance, with no free tile, while paused, when regressed, with invalid intervals and chances, and with an ability on a plant that lacks the capability. `FixedRandom` makes them repeatable, and `mvn test` runs them all.

Summary

Principle	Where it shows	Change it absorbs	Where it is only partly met, and why
SRP	Timing, each effect, planting, regressing and pausing each have one class.	A new timing rule edits only `PeriodicAbility`.	`Crop` holds one Merchant message (Section K).
OCP	A new stage, ability, capability or plantable ground is one new class.	A Pumpkin, a wither ability, a Greenhouse floor: no existing class changes.	New plants are placed in `StackQueueValleyWorld`, the composition root.
LSP	Every `Crop` can be watered and harvested; abilities never check the plant's type.	A new stage works wherever `Plant` is expected.	—

Principle	Where it shows	Change it absorbs	Where it is only partly met, and why
ISP	`Growing`, `Spreading`, `Producing`, `Plantable`, `Regressible` and `PlantAbility` each have one method.	Parsnip carries only `produce()`.	—
DIP	Abilities depend on capabilities, pests on `Regressible`, `Plant` on `PlantAbility`; `RandomUtils` and `AreaFactory` are injected.	A new plant or pest needs no change to the other side.	Each plant creates its own engine `Display`; stages create their next stage directly (Connascence of Name, one outcome each, so a factory would be a parameter nobody varies).

DRY and KISS. Each rule exists once: the timing, the priority loop, planting and the countdown. We added no layer that only one class uses. **Connascence.** Between packages it is only of Name and Type, through `Regressible`, `Plantable` and the capability interfaces. Stronger forms (Position for ability priority, Execution for age before abilities) stay inside one class each.

Jingyu Zhang 37033956 – REQ 3: Infestation and Pest Control

Class diagram: `docs/A3_UML/A3_REQ3_Infestation_UML.drawio`. Feature proposal: `feature-proposal.md`.

Pests move towards plants and eat them. The Farmer fights back with three defences: one pulls pests in, one hunts them, and one pushes them out. Every effect changes the map or an actor, never only prints text.

Class	Abstraction	Effect	Game state it changes
`SlimeSnail`	`Infestation` (`feedOn`)	Sets the plant back and slimes the bare ground around it	Plant stage; ground over an area, restored later
`LocustSwarm`	`Infestation`	Sets back every plant on and around it; may split; leaves when full	Plant stages over an area; spawns and removes actors
`BurrowingMole`	`Infestation`	Sets the plant back and caves the four straight neighbours into Dirt	Plant stage; ground
`PestTrap`	`PestDeterrent` (`deterAround`)	Becomes a trap that pulls pests closer and captures them	Ground; forced movement; removes actors
`LadybugJar`	`PestDeterrent`	Releases up to three ladybugs that eat pests	Spawns actors
`SaltBarrier`	`PestDeterrent`	Salts the bare ground around the Farmer; burns and pushes out pests on it	Ground over an area; hit points; forced movement

Each decision compares rejected and chosen designs, with pros, cons and a future example. Sections H and J list every place a reader may see as a weakness, and say why we chose it.

A. Two interfaces, and one pest turn written once

Why interfaces. Rejected: `Infestation` and `PestDeterrent` as abstract classes.

Pros	Cons
One type fewer on each side.	Java has single inheritance. `Pest` already extends `WanderingActor` (from A2) and `DeterrentItem` extends the engine's `Item`, so neither could also extend an abstract `Infestation` or `PestDeterrent`.

Chosen: both are one-method interfaces. A pest is an actor and an `Infestation` at once, and defences find it through `asCapability(Infestation.class)`, not through the `Pest` class (DIP, ISP). **Foresight:** a cursed scarecrow that is not a `Pest` could implement `Infestation`, and every trap, jar and salt bag would find it unchanged.

The pest turn. Rejected: one `Pest` class with a `PestKind` enum and a `switch` in `feedOn`.

Pros	Cons
KISS: one file holds every pest.	OCP: a new pest edits the switch, and every other switch on the enum.
A reader sees all pest rules side by side.	SRP: one method mixes slime, area damage and cave-ins. This is the "switch statement" smell.
	Each case would need its own fields (meals left only for the swarm), so the class carries fields most pests never use.

Chosen: `Pest` is an abstract class (Template Method). Its turn is written once: if the pest can feed here, it returns an `InfestAction`; otherwise its `SeekPlantBehaviour` moves it. A subclass gives only its landing offsets and its `feedOn`. One hook, `canFeedAt(Location)`, is true when a plant is underfoot. `LocustSwarm` overrides it, because a swarm also feeds on the plants next to it.

Pros	Cons
OCP: a new pest is one new class. `Pest`, `InfestAction` and the defences do not change.	A reader must open two classes to see one pest's full turn.
LSP: the turn never asks which pest it is. The hook replaces `if (this instanceof LocustSwarm)`, which would make the parent know its children.	A subclass could override `canFeedAt` badly, for example return true where `feedOn` finds nothing. `feedOn` then returns an empty report, so the game does not break.

Pros	Cons
DRY: "feed or move" exists once. `Pest` extends the existing `WanderingActor`, so the unconscious check from A2 is reused, not copied.	

Why `feedOn` is not a template. We considered a template on `Pest` where each pest gives only a per-tile step. But the three effects touch different tiles (the area around the swarm, the mole's four straight neighbours, the bare ground around the snail), do different things to them, and only the swarm changes itself (meals, splitting). A template would wrap three different effects without removing repeated code (KISS).

Connascence: `InfestAction` knows a pest only through `Infestation` (Connascence of Name). **Foresight:** an Aphid that hops three tiles and poisons the soil is one class: a new offset table and a new `feedOn`.

B. Movement as data, not as classes

Option	Pros	Cons
Each pest overrides `move()`	Easy to read in one class.	DRY: "find the nearest plant, pick a landing, fall back" is copied three times. A bug fix needs three edits (Connascence of Algorithm).
Three classes (`KnightMove`, `CrawlMove`, `BurrowMove`)	Each move has a name.	Three classes that differ only in a table of numbers. The A2 marker called one class per variant "architectural bloat".
One `SeekPlantBehaviour` with offsets (chosen)	DRY: one copy of the search. OCP: a new movement is one new array. KISS: no new type per pest.	The offsets are a raw `int[][]`, a mild "primitive obsession" smell.

We accept the smell because the array never leaves its pest class, and the behaviour validates and copies it in its constructor. So a pest cannot pass `{1}` or change the table later. The fallback is typed as the engine's `Behaviour`, not `WanderBehaviour` (DIP). The pest lands on a tile and does not walk through the tiles between, so a jumping pest can pass a wall or a salted tile.

Two rules came from running the game. A mole that only moved two tiles could never reach a plant one tile away, so it now moves one or two. A knight jump rarely lands on a plant, so the swarm feeds from the tiles next to it. Unit tests cover both.

Foresight: a grasshopper that leaps three tiles in a straight line is one new offset table; `SeekPlantBehaviour` stays the same.

C. Pests depend on `Regressible`, not on crops

Pests never name a crop. They ask the ground for `Regressible` (from REQ1) with `getGroundAs` and call `regress`.

Pros	Cons
DIP: `game.pests` does not import `game.crops`, so there is no package cycle.	A pest cannot treat one crop differently, for example a snail that prefers Strawberries. That would need a new method on `Regressible`.
OCP: a new plant that can regress is eaten without any change to the pests.	
No `instanceof`: `getGroundAs` is the engine's capability lookup.	

D. One deploy action for all defences

Rejected: three actions, `SetTrapAction`, `OpenJarAction` and `ThrowSaltAction`.

Pros	Cons
Each action can have its own menu text.	DRY: each repeats "remove the item, act, report". If the rule for using an item changed, three classes change (Connascence of Algorithm).

Pros	Cons
	OCP: each new defence also needs a new action.

Chosen: `DeterrentItem` is an abstract item. While the Farmer carries it, it offers one `DeployDeterrentAction`, which calls `deterAround` and removes the item. The action holds a `PestDeterrent`, not a `PestTrap` (DIP). The hook `canDeployAt(Location)` hides the option where it cannot work: a trap is not offered on a plant, so a failed attempt never wastes the item.

Pros	Cons
OCP/DIP: a new defence is one subclass. The action never changes.	The action keeps two fields for one object: the `Item` (the engine's inventory only removes an `Item`) and the `PestDeterrent`. They must be the same object (Connascence of Identity). `DeterrentItem` passes `this` twice in one place, so callers cannot mix them up.
SRP: the item says what it does; the action says how an item is used.	

Rejected fixes for the two fields: a cast breaks type safety; a generic `<T extends Item & PestDeterrent>` adds a type parameter that only removes one field.

Foresight: a Scarecrow that scares pests away for ten turns is one `DeterrentItem` subclass plus one `TemporaryGround` subclass.

E. Finding pests without `instanceof`

The trap, the salt and the ladybug all need to tell a pest from the Farmer, a cow or another ladybug. `PestPresent` is a `TileCondition` (the area filter from A2). It checks `actor.asCapability(Infestation.class)`, which only pests have.

Option	Pros	Cons
`instanceof Pest` in each defence	Short.	A type check is a code smell, and each defence names the pest class. A pest that is not a `Pest` subclass would be missed.
A list of pest classes	Explicit.	Every new pest edits the list (OCP).
`PestPresent` (chosen)	DRY: one check, used in three places. OCP: a new pest is found by every defence at once.	One more small class.

F. Temporary ground

Slime, salt and the trap replace a tile's ground for a while and then put the old ground back.

Rejected: each ground keeps its own counter.

Pros	Cons
No abstract class to learn.	DRY: the same countdown in three classes. A change such as "rain shortens every temporary ground" edits three classes.
	A ground can forget to restore the old ground, a bug the compiler cannot catch.

Chosen: `TemporaryGround` is abstract. Its `tick` is `final`: run `tickEffect` (empty by default), stop if the effect already replaced this ground, then count down and restore the original ground.

Pros	Cons
The order "effect, check, count down" is fixed in one place (Connascence of Execution kept local).	`final` stops a subclass from changing the countdown itself, for example a ground that never expires. Such a ground would extend `Ground` directly.
KISS: `SlimeGround` and `SaltGround` are only constants.	

Package placement. `TemporaryGround` sits in `game.positions`. `SlimeGround` sits in `game.pests`, and `SaltGround` and `TrapGround` in `game.pestcontrol`. If all three were in `game.positions`, that package would import `game.pests`, which already imports `game.positions`: a cycle. Each ground sits beside the feature that creates it (Acyclic Dependencies Principle).

G. The three defences

PestTrap and TrapGround (pull in). Each turn the trap pulls every pest within two tiles one step closer, then captures pests on or next to it, up to two. Grounds tick before actors move. In our first version the trap captured first; a pulled mole then walked away on its own turn, and the trap never caught it. Pulling first fixed this (Connascence of Timing, explained in the class comment and locked in by a test).

A ground's `tick` returns nothing, so the trap cannot hand its report to the engine as an action does.

Option	Pros	Cons
A `static Display`	One shared object.	A static field; the A2 marker penalised statics.
A `private final Display` field, as `Plant` does	Matches REQ1.	Every trap holds a `Display` it rarely uses, and the diagram gains an association with no state.
A `Display` passed into the constructor	Testable.	The `Display` would travel through the item and the action, which have nothing to print.
Make the trap an actor	Uses the engine's normal message path.	A trap is not a creature; it changes the TA-approved design.
A local `Display` when there is news (chosen)	No static state; a dependency, not an association.	Tests check the map, not the printed text.

LadybugJar and Ladybug (hunt). The jar releases up to three ladybugs around the Farmer. A ladybug eats a pest next to it, or wanders. It loses one hit point a turn, so it lives five turns, and the existing `ConsciousActor` removes it at zero (DRY). The cost: lifespan and health are one number. If something healed a ladybug, it would live longer. Nothing in the game does.

SaltBarrier and SaltGround (push out). The bag salts bare Soil and Dirt around the Farmer, burns any pest on those tiles and pushes it out. All of this happens once, inside `deterAround`. After that, `SaltGround` only refuses pests in `canActorEnter` (KISS).

Why an item and a ground for the trap and the salt. This looks like a "parallel hierarchy" smell. But the engine keeps carried things (`Item`) and map tiles (`Ground`) as separate types, and one object cannot be both. The item lives in the inventory; the ground lives on the map with its own timer.

H. Visibility, `final`, `protected` and `static`

A2 feedback said package-private helpers (`CrowSensing`, `CrowNavigation`) can be good encapsulation, but the rationale must say so. Here is each choice in REQ3.

Choice	Where	Why	Cost and when we would change it
Every class is `public`	All 20 REQ3 types, including the small `PestPresent`, `InfestAction` and `EatPestAction`	A3 is the last assignment, so package-private would be safe today. We still chose `public`, because extensibility is a design goal and other features already live in other packages. A defence in another package, for example a frog in `game.actors`, should reuse `PestPresent` and `EatPestAction` without moving them.	A wider public API. We considered package-private for `InfestAction` (only `Pest` uses it) but chose one rule for the whole feature, so a reader never wonders whether a hidden class was forgotten.

Choice	Where	Why	Cost and when we would change it
Every field is `private`	All classes	Implementation details stay inside (A2 feedback). Subclasses use hooks, not fields.	`SlimeSnail`, `BurrowingMole` and `LocustSwarm` each keep their own `AreaFactory`, although `Pest` also receives it. A `protected` getter on `Pest` would remove three fields but expose a tool that `Pest` itself does not use.
Constants are `private static final`	e.g. `LocustSwarm.SPLIT_CHANCE_PERCENT`, `SlimeSnail.HIT_POINTS`	A fixed rule belongs to the class, not to one object. A name such as `SPLIT_CHANCE_PERCENT` explains the number, so there are no magic numbers. Java does not allow instance fields in `super(...)`, so names, characters and hit points must be static to be named.	A test cannot read them, so tests repeat the number.
No static methods, no `public` constants and no mutable static fields	—	This is what the A2 marker penalised: static methods are "function thinking", and `public` constants leak details. A `private static final` constant does neither: it cannot change and is not seen outside the class.	—
`protected` constructors	`Pest`, `DeterrentItem`, `TemporaryGround`	They are abstract; only subclasses construct them.	—
`protected` hooks	`canFeedAt`, `canDeployAt`, `tickEffect`	Subclasses must override them; callers must not skip the template.	Classes in the same package can also call them. No class does.
`protected final`	`TemporaryGround.revertAt`	A subclass may end its ground early (the trap after two captures) but cannot change how it ends.	—
`final` method	`TemporaryGround.tick`	Fixes the order of the countdown (Section F).	See Section F.
Classes are not `final`	e.g. `LadybugJar`, `SaltGround`	Open for extension (OCP), e.g. a larger jar.	A subclass could break a rule. Private fields and the final `tick` limit the damage.
`private` constructor	`LocustSwarm(int meals, ...)`	Only a split may create a smaller swarm. The public constructor always gives four meals, so a half-empty swarm cannot be built by mistake.	A future "hungry swarm" difficulty would need a new public constructor.
`private` helpers	e.g. `pullPests`, `splitAt`, `copyOf`	They are steps, not contracts. Making them `protected` would freeze their signatures for every subclass.	A subclass that wants a different pull must override `tickEffect`.
Non-final fields	`mealsLeft`, `capturesLeft`, `remainingTurns`	These values change every turn.	—

Explicit modifiers, and why we keep `protected`. Two pieces of advice pull in different directions. The Week 9 lecture and our tutor suggest a narrow API: keep a class package-private when no other package needs it, and try not to use `protected`, because in Java it also opens a member to every class in the same package. But Java has no keyword for package-private: it means writing no modifier at all, and the A2 marker penalised exactly that ("missing modifier" on `CrowSensing` and `ForageableItems`).

Option	Pros	Cons
Package-private classes and hooks (no modifier) where every user is in the same package	The narrowest API; fewer dependencies across packages.	A reader cannot tell a deliberate choice from a forgotten modifier, which A2 penalised. A pest or defence in another package could not extend `Pest` or `DeterrentItem`.
Replace the template hooks with strategy objects, so no `protected` is needed	No `protected` at all.	A new interface and class for each hook (`canFeedAt`, `canDeployAt`, `tickEffect`) only to avoid a modifier: needless complexity (KISS).
Explicit `public` classes and `protected` hooks (chosen)	Every class and member states its access, so nothing looks forgotten. `protected` is used only for constructors of abstract classes and for hooks that subclasses override; no field is `protected`.	A wider API than strictly needed, and classes in the same package could call a hook. None does, and the template methods that call the hooks are `final` or fixed in the parent, so a hook cannot skip the rules.

I. Changes to the earlier design

REQ3 is new in A3. The only earlier class it changes is `StackQueueValleyWorld`, which places the pests and gives the Farmer the defences.

Statics replaced by injection. REQ3's first version called the A2 static helpers `Area.within`, `Area.adjacentTo` and `RandomUtils.rollPercent`. The A2 marker called static helpers "function thinking". Following that feedback, the team now creates one `RandomUtils` and one `AreaFactory` in `StackQueueValleyWorld.initialise`, and every REQ3 class that needs them receives them through its constructor.

Pros	Cons
DIP: a pest depends on the objects it is given, not on global classes. Tests pass `FixedRandom` directly.	Every pest and defence constructor gained one or two parameters, and `Pest` must pass them on (a "tramp data" smell).
No hidden global state, so two tests cannot affect each other.	Connascence of Position in the constructors. Each has at most two such parameters, so the order is easy to check.

Display character. `SaltBarrier` first used `k`, which clashed with the A2 `CauliflowerItem`. Earlier characters are fixed, so the new class changed to `f`.

J. Known minor violations and why we accept them

Violation or cost	Why it is acceptable
`SlimeSnail` and `SaltBarrier` both ask "is this tile bare Soil or Dirt?" (two lines each). DRY is not fully met.	Rule of three: two short copies are cheaper to read than a new class. A third use would be the signal to extract a `BareFarmGround` condition.
`LadybugJar` catches `GameEngineException` and moves on. An empty catch is a smell.	The tile is checked with `canActorEnter` first; the catch only covers a tile taken in between. A comment says so.
`Ladybug`, `LadybugJar` and `SaltBarrier` use `getExits()`, while other classes use `AreaFactory`.	`getExits()` is the engine's own list of neighbours and needs no extra dependency. `AreaFactory` is used where we filter or search an area.
`TrapGround` moves actors, which is more than most grounds do.	Acting on the pests around it is the trap's whole job. Only the trap does this each turn, and the logic stays in one class.

Violation or cost	Why it is acceptable
Pests (except the swarm) never leave on their own.	On purpose: the defences are the counter to the pests. A meal limit could be added in `feedOn` only.

Summary

Principle	Where it shows	Change it absorbs	Where it is only partly met, and why
SRP	Moving (`SeekPlantBehaviour`), the turn (`Pest`), finding pests (`PestPresent`), using an item (`DeployDeterrentAction`) and timing ground (`TemporaryGround`) each have one class.	A new landing rule edits `SeekPlantBehaviour` only; a new countdown rule edits `TemporaryGround` only.	`TrapGround` pulls, captures and prints. `LocustSwarm` feeds, splits and leaves. Each is one rule from the brief, and splitting them would add classes used once.
OCP	A new pest, defence or temporary ground is one new class.	An Aphid, a Scarecrow or a Frost ground: no existing class changes.	The new class must still be placed in `StackQueueValleyWorld`. That is the composition root, so this is expected.
LSP	Every pest works wherever `Pest` or `Infestation` is expected; every defence wherever `PestDeterrent` is. Hooks replace type checks.	A new pest is fed, found and trapped with no new branch.	`canFeedAt` and `feedOn` must agree. If they do not, `feedOn` returns an empty report, so the game stays correct. `final tick` stops a ground from breaking the countdown.
ISP	`Infestation`, `PestDeterrent` and `Regressible` each have one method. Defences never see a pest's movement or health.	A pest that moves differently does not touch any defence.	`PestPresent` uses `Infestation` only to recognise a pest and never calls `feedOn`. This is not a marker interface: `InfestAction` calls `feedOn`.
DIP	`DeployDeterrentAction` depends on `PestDeterrent`, pests on `Regressible`, `SeekPlantBehaviour` on `Behaviour`; `RandomUtils` and `AreaFactory` are injected.	A new crop, defence or fallback behaviour needs no change to the classes that use it.	`Pest` creates `SeekPlantBehaviour` and `WanderBehaviour`; `SlimeSnail` creates `SlimeGround`; `LadybugJar` creates `Ladybug`. Each has one outcome, so an injected factory would be a parameter nobody varies (YAGNI).

DRY and KISS. Each rule exists once: the pest turn, the landing search, finding a pest and the countdown. We added no layer that only one class uses. **Connascence.** Between packages it is only of Name and Type. Stronger forms (Timing, Identity, Execution) stay inside one class each, with a comment and a test.

Testing. Every class has normal, edge and invalid tests. `FixedRandom` makes them repeatable, and `mvn test` runs them all.

Yuanshu Liu 36226173 – REQ 4: Enchanted Tools

Feature proposal: `feature-proposal.md`.

The Farmer can enchant a carried Hoe, Watering Can or Pickaxe with one of three Enchantments. The next time the tool does its normal job, two effects fire in the same turn: the tool's own spread effect, which is the same whichever Enchantment is attached, and the Enchantment's effect on every tile the spread touched. Then the Enchantment is spent. An unenchanted tool works exactly as in A1 and A2.

The brief asks for two abstractions with three new classes and up to three retrofitted ones. Every effect changes the map or an actor, never only prints text.

Class	Abstraction	New or retrofitted	Effect	Game state it changes
`Hoe`	`Enchantable`	Retrofitted (A1)	Also tills every adjacent tillable tile	Ground over an area
`WateringCan`	`Enchantable`	Retrofitted (A1)	Also waters one random adjacent waterable tile	Ground or crop watering count
`Pickaxe`	`Enchantable`	Retrofitted (A2)	40% chance to also break one random adjacent breakable tile	Ground; drops items
`BlazeEnchantment`	`Enchantment`	New	Lights a `Fire` on every affected tile	Spawns items; burns actors
`BountyEnchantment`	`Enchantment`	New	Gives every plant on an affected tile an extra growth attempt	Plant stage
`ShatterEnchantment`	`Enchantment`	New	Stuns every actor on an affected tile for 2 turns	Status on actors

Each decision compares rejected and chosen designs, with pros, cons and a future example. Sections H and J list every place a reader may see as a weakness, and say why we chose it.

A. Finding an enchanted tool without `instanceof`

`CapabilityToolAction` (tilling and watering) and `BreakAction` must fire the Enchantment of the tool they just used.

Rejected: check the tool's class, for example `if (tool instanceof Hoe)`, and read its enchantment directly.

Pros	Cons
Short; no new interface.	OCP: every new enchantable tool adds a branch to two actions.
	Connascence of Type between two actions and every tool class.

Chosen: a capability interface, `Enchantable`, with `enchant(Enchantment)` and `triggerEnchantment(Actor, Location)`.

Pros	Cons
DIP: both actions depend only on `Enchantable`.	One more interface to read.
OCP: a new enchantable tool needs no change to either action.	
The tool decides whether it has anything to fire (encapsulation).	

This is the same idiom the codebase already uses for `Tillable`, `Waterable`, `Breakable` and `Harvestable`, so a reader needs no new pattern. **Foresight:** an enchantable Scythe implements `Enchantable` through the base class in Section B, and the actions stay the same.

B. Sharing "can be enchanted" in one base class

All three tools must store the Enchantments they offer, offer one `EnchantAction` per Enchantment, and hold the one currently attached. None of this depends on the tool.

Rejected: each tool implements `Enchantable` on its own.

Pros	Cons
No new abstract class.	DRY: the same storage and menu code three times ("duplicated code" smell).
	A change to how `EnchantAction`s are offered needs three edits ("divergent change" risk).

Chosen: `AbstractEnchantableTool extends Item implements Enchantable`. It owns the offered Enchantments, the attached one (through `EnchantmentSlot`, Section C), `enchant()` and the menu. `Hoe`, `WateringCan` and `Pickaxe` extend it and write only `triggerEnchantment()`.

Pros	Cons
DRY: written once.	A tool that should offer only some Enchantments must override `allowableActions()`. No tool needs this today (YAGNI).
SRP: each tool changes only when its own spread changes.	

An interface default method cannot hold a field, so sharing the stored Enchantment needs a class. This is a language limit, not a style choice. **Foresight:** a fourth enchantable tool writes only its `triggerEnchantment()`.

C. Storing, firing and clearing one Enchantment

Firing is the same for every tool: if an Enchantment is attached, apply it to the affected tiles, then clear it. What differs is which tiles the tool's own spread touches.

Rejected: write the store, fire and clear steps inside each tool's `triggerEnchantment()`.

Pros	Cons
One class fewer.	SRP: one method mixes the bookkeeping with the tool's spread.
	DRY: the "check, apply, clear" steps three times.

Chosen: `EnchantmentSlot`, a small helper that holds at most one Enchantment and offers `enchant()`, `isEmpty()` and `fire(tiles, activator, origin)`, which applies the effect and clears itself.

Pros	Cons
Each tool reads as "build my list of tiles, hand it to the slot".	One more step to follow when tracing a turn: tool, then slot, then Enchantment.
The same role as other helpers in the codebase, such as `StatisticAdjuster` and `MerchantContext`.	

D. Finding which tool was used

`CapabilityToolAction` and `BreakAction` both need the same lookup after their own effect: find the tool instance just used among the actor's `Enchantable` items, and fire it.

Rejected: write the lookup loop inside each action.

Pros	Cons
No new class.	DRY: the same loop in two actions. If the lookup changes, both copies must change together (Connascence of Algorithm).

Chosen: `EnchantmentTrigger.fire(Actor, Object usedTool, Location)`, used by both actions. In the first version only `CapabilityToolAction` needed the lookup. When `BreakAction` needed the same loop, we moved it into one class. So the class was created for a real second user, not "just in case".

Pros	Cons
One place knows how to go from a tool in hand to its `Enchantable` view.	The `usedTool` parameter is typed `Object` (Section J).
A future action that uses a tool reuses it with no new code.	

E. Why `==` is used, and why it is not a hidden `instanceof`

The engine's `Inventory.getItemsAs(Class)` returns views narrowed to one capability. A tool held as a `TillingTool` cannot be asked for a second capability, and the engine cannot be changed. So `EnchantmentTrigger` takes a second view, `getItemsAs(Enchantable.class)`, and looks for the entry that is the same object as the tool in hand (`enchantable == usedTool`).

This is Connascence of Identity, the strongest kind, because the two references must be the same object. We accept it because:

- it is the real question: "is this the tool used this turn?";
- it is fully local: both references are found and compared inside one method, in one turn, and never stored;
- `instanceof` plus a cast would ask a different question ("what class is this?") and bring back the Connascence of Type that Section A removed.

F. Reusing existing classes for the Enchantment effects

Enchantment	Reuses	Instead of
`BlazeEnchantment`	The existing `Fire` item	A new burning mechanism
`ShatterEnchantment`	The existing `StunnedStatus`	A new stun status
`BountyEnchantment`	REQ1's public `Plant.attemptGrowth()`, which tries the plant's abilities without ageing it or printing	A copy of the plant's ability loop

Pros	Cons
DRY: each effect already exists and is already tested.	`game.enchantments` depends on `game.items`, `game.status` and `game.plants`. These are ordinary dependencies on existing classes and form no cycle.

G. The empty-string contract

`Enchantable.triggerEnchantment`, `EnchantmentSlot.fire` and `Enchantment.applyTo` never return `null`. They return `""` when nothing fired, the same rule REQ1's abilities use.

Option	Pros	Cons
Return `null` for "nothing happened"	Common in Java.	Connascence of Meaning: every caller must remember what `null` means, and a forgotten check crashes.
Return `""` (chosen)	Every caller simply joins the strings. One rule for the whole codebase.	A reader must know that `""` means "nothing happened"; the Javadoc says so.

Every message still names a location: when there are no affected tiles, each Enchantment reports "near" the origin, and both actions start their message with "At" plus the location.

H. Visibility, `final`, `protected` and `static`

Choice	Where	Why	Cost
Every class is `public`	All REQ4 classes	Another package can reuse them, for example a new action that uses `EnchantmentTrigger`.	A wider public API.
Every field is `private`	All classes	Subclasses use `enchantmentSlot()`, not the field.	—
`protected` accessor	`AbstractEnchantableTool.enchantmentSlot()`	Only the tools need the slot to fire it; callers use `Enchantable`.	Classes in `game.items.tools` could also call it. None does.

Choice	Where	Why	Cost
`protected` constructor	`AbstractEnchantableTool`	It is abstract; only tools construct it.	—
`final` classes	`EnchantmentSlot`, `EnchantmentTrigger`	Small helpers used by composition, not by inheritance. A subclass could break "fire once, then clear".	A variant slot, such as one holding two Enchantments, would be a new class.
Constants are `private static final`	e.g. `CHAIN_BREAK_CHANCE_PERCENT`, `STUN_DURATION_TURNS`	A name explains each number; no public constants (A2 feedback).	Tests repeat the numbers.
No static methods	—	A2 feedback penalised static methods.	—

I. Changes to the earlier design

Change	Why	Cost
`Hoe`, `WateringCan` and `Pickaxe` extend `AbstractEnchantableTool` instead of `Item`.	They are the three retrofitted classes (Section B).	An unenchanted tool behaves as before; its menu gains the enchant options.
`CapabilityToolAction` and `BreakAction` call `EnchantmentTrigger` after their own effect.	So every tool use can fire its Enchantment (Section D).	One more call in each action.
`TillingTool` and `WateringTool` declare `canTill(Tillable)` and `canWater(Waterable)`.	A2 feedback: empty marker interfaces are not good OO design.	Every tool returns `true` today.
`Enchantment` and `EnchantAction` live in `game.items.tools`.	Keeping them there removes a package cycle between `game.items.tools` and `game.enchantments`.	The interface sits beside the tools, not beside its implementations.
The tools receive their Enchantments through the constructor.	No hard-coded list inside the tools; `StackQueueValleyWorld` decides what is offered.	One more constructor parameter.

J. Known minor violations and why we accept them

Violation or cost	Why it is acceptable
`EnchantmentTrigger.fire` takes `Object usedTool`, so a wrong argument fails silently (no match, `""`) instead of at compile time.	The two callers hold tools of different types (`T extends HasStaminaCost` and `BreakingTool`), and `HasStaminaCost` does not say the tool is an `Item`. A narrower type needs a new shared interface or a new generic bound on `CapabilityToolAction`, only to tighten one parameter (YAGNI).
The three `triggerEnchantment()` bodies share a shape: check the slot, build a tile list, fire, join the messages.	Their spreads differ in kind: every adjacent tile, one random tile, or one random tile behind a chance roll. What they share, the slot logic, is already in `EnchantmentSlot`. With three different shapes, a common template would be a guess (rule of three).
`game.enchantments` depends on three other packages (Section F).	Reuse is the point; there is no cycle.

Summary

Principle	Where it shows	Change it absorbs	Where it is only partly met, and why
SRP	Offering (`AbstractEnchantableTool`), storing and firing (`EnchantmentSlot`), finding the used tool (`EnchantmentTrigger`), each spread (the tools) and each effect (the Enchantments) each have one class.	A new firing rule edits only `EnchantmentSlot`.	—
OCP	A new tool or Enchantment is one new class.	A Scythe or a Frost Enchantment: no existing class changes.	New Enchantments are listed in `StackQueueValleyWorld`, the composition root.
LSP	Any `Enchantable` works in both actions; any `Enchantment` works in any slot.	A new tool fires with no new branch.	—
ISP	`Enchantable` has two methods and `Enchantment` one; the actions see only `Enchantable`.	—	—
DIP	The actions depend on `Enchantable`, the tools on `Enchantment`; `RandomUtils` and `AreaFactory` are injected.	A new Enchantment needs no change to the tools.	`EnchantmentTrigger` is created with `new` in each action; it has no state and one possible behaviour.

DRY and KISS. The menu, the slot and the lookup each exist once, and the Enchantments reuse `Fire`, `StunnedStatus` and `Plant.attemptGrowth`. **Connascence.** Between packages it is only of Name and Type. The one Connascence of Identity (`==`) stays inside one method.

Testing. `HoeTest`, `WateringCanTest`, `PickaxeTest`, `EnchantmentSlotTest` and one test class per Enchantment cover each tool's own spread, tools with and without an Enchantment, Enchantments with and without affected tiles, and that an Enchantment is spent after one use. `FixedRandom` makes them repeatable, and `mvn test` runs them all.

Jingyu Zhang 37033956, Yuanshu Liu 36226173, Poyee Tan 36641162 – REQ 5: The Wanderer (API, HD)

Feature proposal: `feature-proposal.md` (thresholds, request and response schema).

A Wanderer roams the farm, and the real weather where it stands decides what it does. Its tile is turned into a real latitude and longitude, so every step changes the place it asks about. Every 5 turns it reacts to the current weather. A Farmer next to it can ask for tomorrow's forecast, and an omen follows. Every effect changes the map or an actor, never only prints text.

Class	Abstraction	Effect	Game state it changes
`StormShelterReaction`	`WeatherReaction`	Builds a shelter (`StormShelterGround`) on every free adjacent tile; stronger wind lasts longer	Ground over an area, restored later
`RainDanceReaction`	`WeatherReaction`	Tills bare Dirt and gives crops a free watering in an area that grows with the rainfall	Ground; crop watering counts
`CampfireReaction`	`WeatherReaction`	Lights `Fire` on 1 to 8 adjacent tiles; colder means more tiles	Spawns items; burns actors
`FrostOmen`	`ForecastOmen`	Pauses the growth of every plant within 2 tiles	Status on plants
`SowingOmen`	`ForecastOmen`	Plants a Strawberry Seedling on every bare Soil tile within 2 tiles	Spawns plants
`DroughtOmen`	`ForecastOmen`	Drains the asker's stamina for a few turns	Status on an actor

`Wanderer`, `ReactToWeatherAction` and `AskForecastAction` are the higher-level classes. They depend only on `WeatherReaction` and `ForecastOmen` and never name a concrete reaction or omen.

Each decision compares rejected and chosen designs, with pros, cons and a future example. Sections I and K list every place a reader may see as a weakness, and say why we chose it.

A. Keeping the game independent of the weather API

Rejected: `Wanderer` and `AskForecastAction` build the URL, send the request and read the JSON themselves.

Pros	Cons
One interface and one class fewer.	DIP: game classes would depend on `HttpClient` and on OpenWeatherMap's JSON shape.
	Every test of the Wanderer would need the network or a fragile HTTP mock.
	Changing the provider would mean editing `Wanderer`.

Chosen: a `WeatherSource` interface with `currentWeatherAt(lat, lon)` and `tomorrowWeatherAt(lat, lon)`, both returning `Optional<WeatherReading>`. `OpenWeatherClient` is the real one and is injected through the constructor. Tests inject `FixedWeatherSource`.

Pros	Cons
DIP: game classes depend on an interface the game owns.	One more interface and one more class.
No REQ5 test uses the network.	
OCP: another provider, or a cached one, needs no change to `Wanderer`, `AskForecastAction` or `WeatherInterpreter`.	

The package `game.weather` imports nothing from the rest of the game, so it could be reused by any feature.
Foresight: a weather source that replays saved readings for a demo is one new class.

B. Choosing which reaction today's weather triggers

Option	Pros	Cons
An if-chain inside `Wanderer.playConsciousTurn` that also performs each effect	No extra class.	SRP: the actor would also hold every weather rule. OCP: a new reaction edits `Wanderer`. The bare numbers would need a reader to guess their meaning (Connascence of Meaning).
A configurable rule engine: a list of (condition, effect) pairs, perhaps read from a file	Rules could change without recompiling.	The six rules are fixed by the proposal. A generic list is harder to read than a few `if`s, and solves a problem we do not have (YAGNI, KISS).
`WeatherInterpreter` with `reactionFor(reading)` (chosen)	`Wanderer` sees only `WeatherReaction`. The thresholds are named constants that exist only here. The priority order (wind, then rain, then cold) can be read in a few lines.	One more class to learn.

`WeatherInterpreter` is the only class that creates the six concrete classes. It works as a simple factory. `CampfireReaction` and `FrostOmen` size their effect by how far the temperature is below the threshold that chose them, so `WeatherInterpreter` passes that threshold into their constructors. Our first version repeated 10°C and 2°C as constants in those two classes (Connascence of Value across classes): changing a threshold in one place would have left the effect sized against the old number. Passing it in leaves one source for each number. **Foresight:** a `HeatwaveReaction` above 35°C is one new class and one new `if` in `WeatherInterpreter`; `Wanderer` and the other reactions do not change.

C. Two interfaces, not one

Rejected: one interface for both sides, with `reactAround(Location)` and `bringUpon(Actor, Location)`.

Pros	Cons
One interface to learn.	ISP: every reaction would carry an omen method it cannot use, and every omen a reaction method. The unused bodies would do nothing or throw, so a caller could not trust every implementation (LSP risk).

Chosen: `WeatherReaction` (`reactAround(Location)`) and `ForecastOmen` (`bringUpon(Actor, Location)`). Both are chosen in the same `WeatherInterpreter`, because "which effect does this reading trigger" is one job for today's and tomorrow's weather.

Pros	Cons
ISP: each class implements only the method it needs.	Two interfaces that look alike.
DIP: `AskForecastAction` depends only on `ForecastOmen`; `Wanderer` and `ReactToWeatherAction` only on `WeatherReaction`.	
DRY: `isRainy` and the condition names are shared by both methods inside one class.	

D. When the sky cannot be read

With no key, no network or a bad reply, there is no reading.

Option	Pros	Cons
Throw a checked exception	Conventional Java.	Every caller must remember the `try/catch` in the right place.
Return `null`	Short.	A forgotten check crashes the game. The rule "`null` means no reading" is invisible to the compiler (Connascence of Meaning).

Option	Pros	Cons
Return `Optional<WeatherReading>` (chosen)	"No reading" is part of the return type, so a caller sees it.	Call sites use `isEmpty()` and `get()`.

When there is no reading, `Wanderer` wraps its normal wandering in `UnreadableSkyAction`, which reports "The sky is unreadable, and the Wanderer only wanders." `AskForecastAction` reports that tomorrow's weather is unreadable. The game never stops, and the player always sees what happened. We use `Optional` only with `isEmpty()` and `get()`, with no chained calls.

E. Turning the Wanderer's tile into a real place

Rejected: always ask about one fixed place, such as Melbourne.

Pros	Cons
Trivial.	Breaks the brief's rule that the request cannot be a static URL. The game state would not affect the query.

Chosen: `MapGeography` turns a tile's (x, y) into a latitude and longitude from an origin (Melbourne, -37.81, 144.96) and a scale (2 degrees per tile). `Wanderer` and `AskForecastAction` ask it about the tile the Wanderer stands on, so every move changes the query. Latitude stops at the poles and longitude wraps past 180, so any tile gives a valid place.

Pros	Cons
The query is driven by the game state.	One more small class.
SRP: one class knows how map tiles relate to the real world. Moving the farm changes only the constructor arguments.	
`latitudeOf` and `longitudeOf` name the two numbers, so callers cannot swap them (less Connascence of Position).	

F. Keeping the API key out of the code

Rejected: `OpenWeatherClient` calls `System.getenv("OPENWEATHER_API_KEY")` itself.

Pros	Cons
One class fewer.	SRP: the client would also decide where secrets live.
	A test could not control the key without changing the real environment.

Chosen: `ApiKeyLoader` takes the key's name, an environment map and a `.env` path. It checks the environment first, then a `NAME=value` line in `.env`, and returns `Optional<String>`. The client receives a loader through its constructor.

Pros	Cons
SRP: the loader finds the key; the client calls the API.	One more class.
A test passes a plain map and a temporary file.	
Security: the key is never in the source, and `.env` is in `.gitignore`.	

G. Reading the reply

Rejected: read the JSON inside `OpenWeatherClient`.

Pros	Cons
One class fewer.	SRP: the client would both talk to the network and know the JSON layout. A layout change and a network change would edit the same class.

Chosen: `OpenWeatherParser` turns a reply into a `WeatherReading`. The weather condition and the temperature are required; wind and rain default to 0, because OpenWeatherMap leaves out `rain` on dry days. For the

forecast it reads the last of nine 3-hour steps (about 24 hours ahead) and divides `rain.3h` by 3 to get an hourly amount. A reply missing a required field gives no reading. `WeatherReading` is a small value object: its fields are `private final`, and its constructor rejects a blank condition and negative wind or rain.

Pros	Cons
The parser is tested with saved JSON text, with no network.	Gson is an extra library, added through Maven.
A bad reading cannot exist, so the reactions need no checks of their own.	

H. Letting the Farmer ask about tomorrow

Rejected: give `Farmer` a method that searches the map for a Wanderer and asks it.

Pros	Cons
No new action.	"Feature envy" smell: the data (`WeatherSource`, `MapGeography`, `WeatherInterpreter`) belongs to the Wanderer, not the Farmer.
	Any other actor that wants to ask would copy the same code (DRY).

Chosen: `Wanderer.allowableActions` offers an `AskForecastAction` to whoever stands next to it. This is the engine's normal way for one actor to offer an action to another. `ReactToWeatherAction` and `AskForecastAction` only pass the turn on to the reaction or omen they hold.

Pros	Cons
Reuses the engine's protocol, like every other interactive entity.	Any adjacent actor is offered the action (Section K).
SRP: each action has one job, and neither names a concrete class.	

The engine only asks an actor that is on the map for its actions, so `AskForecastAction` never needs to check that the Wanderer is still there.

I. Visibility, `final`, `protected` and `static`

Choice	Where	Why	Cost
Every class is `public`	All REQ5 classes	`game.weather` is meant to be reused by other features.	A wider public API.
Every field is `private`, and `final` where it never changes	All classes; `WeatherReading` is fully immutable	A reading is shared by the interpreter and the effect it creates, so it must not change.	`Wanderer.turnsSinceLastReaction` is the only changing field.
`protected` method	`OpenWeatherClient.fetch(URI)`	The one seam a test overrides to return saved replies without the network.	A subclass could replace the network call. The class comment says it exists for tests.
`private` helpers	e.g. `OpenWeatherParser.readingFrom`, `WeatherInterpreter.isRainy`	They are steps, not contracts.	—
Constants are `private static final`	e.g. `STORM_WIND_THRESHOLD_MPS`, `TIMEOUT`	A name explains each number; no public constants (A2 feedback).	Tests repeat the numbers.
No static methods	—	A2 feedback penalised static methods.	—

J. Changes to the earlier design

REQ5 is new in A3. It builds on earlier classes without changing them, except `StackQueueValleyWorld`.

Earlier class	How REQ5 uses it
`WanderingActor` (A2)	`Wanderer` extends it, so the unconscious check and wandering are reused.
`TemporaryGround` (REQ3)	`StormShelterGround` extends it, so the shelter restores the old ground with no new countdown code.
`Fire` (A2)	`CampfireReaction` lights the same `Fire` item.
`GrowthPausedStatus` (REQ1), `TimedStatus` (A2)	`FrostOmen` pauses plants; `DroughtWearinessStatus` reuses the countdown.
`Plantable`, `StrawberrySeedling` (REQ1)	`SowingOmen` plants through the same rule as everything else.
`StackQueueValleyWorld`	Builds the client, the key loader and `MapGeography`, and places one Wanderer.

K. Known minor violations and why we accept them

Violation or cost	Why it is acceptable
The Wanderer can be burnt by its own campfire if it later walks onto a burning tile.	On purpose: `Fire` burns any actor and never asks who lit it, so it needs no type check. A Wanderer that stays somewhere cold pays for its fires, and it has 50 hit points.
`OpenWeatherClient` creates its own `HttpClient`, a small DIP gap.	Tests replace `fetch`, not the `HttpClient`, so injecting it would add a parameter nobody varies.
The weather condition is a `String` such as `"Rain"`, so the code depends on OpenWeatherMap's names (Connascence of Meaning).	The names are named constants in `WeatherInterpreter`, and the parser is the only class that reads them from JSON.
`AskForecastAction` holds the same three collaborators as `Wanderer` (a small "data clump").	They are passed once, in `allowableActions`. A holder class for three fields would be a class used once.
Any actor next to the Wanderer is offered `AskForecastAction`, not only the Farmer.	Only the player picks menu actions; other actors never choose it.
`WeatherInterpreter` names all six concrete classes, so a new reaction edits it. OCP is only partly met.	It is the one factory point (Section B). Everything else stays closed.
`ApiKeyLoader` and `OpenWeatherParser` turn errors into an empty result.	The game must keep running without weather. Tests cover the missing key, the missing file and the broken JSON.

Summary

Principle	Where it shows	Change it absorbs	Where it is only partly met, and why
SRP	The API call (`OpenWeatherClient`), the JSON (`OpenWeatherParser`), the key (`ApiKeyLoader`), the place (`MapGeography`), the rules (`WeatherInterpreter`) and each effect each have one class.	A new JSON layout edits only the parser.	—
OCP	A new reaction, omen or weather provider is one new class.	A Heatwave reaction or a cached weather source.	`WeatherInterpreter` gains one `if` per new rule (Section K).
LSP	Any `WeatherReaction` or `ForecastOmen` works in its action; any `WeatherSource` works in the Wanderer.	`FixedWeatherSource` replaces the real client in every test.	—
ISP	`WeatherReaction` and `ForecastOmen` each have one method; `WeatherSource` has the two the Wanderer needs.	—	—

Principle	Where it shows	Change it absorbs	Where it is only partly met, and why
DIP	`Wanderer` depends on `WeatherSource` and `WeatherReaction`; `AskForecastAction` on `ForecastOmen`; the client on `ApiKeyLoader`.	A different provider needs no change to the game.	`OpenWeatherClient` creates its own `HttpClient` (Section K).

DRY and KISS. The rules live in one interpreter, the countdowns reuse `TemporaryGround` and `TimedStatus`, and the effects reuse `Fire`, `GrowthPausedStatus` and `Plantable`. Six fixed rules are a few `if`s, not a rule engine.
Connascence. Between packages it is only of Name and Type, through `WeatherSource`, `WeatherReaction` and `ForecastOmen`. Each threshold exists once, in `WeatherInterpreter`, which passes the cold and frost thresholds to the classes that size their effect by them.

Testing. Each reaction, omen, action and API-layer class has its own test class, for example `WeatherInterpreterTest`, `WandererTest`, `OpenWeatherParserTest` and `ApiKeyLoaderTest`. They cover readings that meet or miss each rule, the priority between rules, and invalid input such as a missing key, broken JSON and negative wind. No test uses the network: `FixedWeatherSource` replaces the client, and `OpenWeatherClientTest` overrides `fetch`. `mvn test` runs them all. Separately, `WandererStressHarness` is a manual tool, not part of `mvn test`: it plays many seeded games through the real classes with fake weather to look for rare crashes.